

Shuaipeng (Frank) Dong

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[Personal Website](#) • [Google Scholar](#) • [LinkedIn](#) • [GitHub](#)

RESEARCH INTERESTS

My research interests are in computer vision, continual learning, and knowledge distillation, with a focus on class-incremental object detection, YOLO-World adaptation, and stable teacher-student distillation for memory-efficient lifelong visual learning.

EDUCATION

The Pennsylvania State University University Park, PA

B.S. in Computer Science • B.S. in Mathematics (Systems Analysis) May 2023

Cumulative GPA: 3.55 / 4.00 • Dean's List (multiple semesters) • College Honors Student

PUBLICATIONS & PATENTS

Conference Publications

[1] S. Dong and D. Whitefield, "Video Bandwidth Prediction by GEO Satellite Terminals for QoE," 41st International Communications Satellite Systems Conference (ICSSC), vol. 2024, pp. 26-31, 2024.

[2] S. Dong, "Predicting Invasive Ductal Carcinoma Using a Deep Convolutional Neural Network," in Proc. 3rd International Conference on Artificial Intelligence and Education (ICAIE), pp. 191-196, 2022.

Patents

[3] S. Dong, "Bandwidth Prediction Using Machine Learning," U.S. Patent Application No. 19/081,355, filed Mar. 2025 (pending).

RESEARCH EXPERIENCE

Knowledge Distillation for Class-Incremental Object Detection 2026 - Present

Independent Research • Computer Vision / Continual Learning

Reproducing and benchmarking continual-learning and incremental-detection baselines, including LwF, iCaRL, ERD, and YOLO-IOD, to analyze degradation of old-class performance under sequential class addition.

Deep Convolutional Neural Networks for Medical Image Classification 2022

Undergraduate Research • The Pennsylvania State University

Investigated whether a parameter-efficient convolutional design could detect invasive ductal carcinoma (IDC) from histopathology while remaining robust to the severe class imbalance characteristic of medical-imaging data.

Designed a depth-wise separable CNN in TensorFlow/Keras for IDC

classification on 50x50 histopathology patches, coupled with a reproducible preprocessing, class re-balancing, training, and evaluation pipeline.

Demonstrated that the separable-convolution architecture matched standard convolutional baselines at substantially lower parameter count and computational cost, evidencing a favorable accuracy-efficiency trade-off on imbalanced data.

First-author publication at ICAIE 2022.

PROFESSIONAL & APPLIED RESEARCH EXPERIENCE

Hughes Network Systems June 2023 - Dec. 2025

Member of Technical Staff II, Software Engineering • Germantown, MD

Framed satellite rain-fade prediction as a spatiotemporal learning problem and designed a deep 3D-CNN over highly imbalanced radar-imaging and telemetry time-series, achieving >90% classification accuracy with high precision and recall and improving downstream link reliability.

Hypothesized that targeted synthetic augmentation would counter severe class imbalance; constructed customized synthetic time-series with tailored augmentation that improved F1 and recall by more than 5% and reduced performance variance across weather regimes.

Developed an LSTM bandwidth-prediction model with a custom loss over streaming TCP-packet data, yielding a peer-reviewed publication (ICSSC 2024) and a pending U.S. patent.

Built end-to-end ML pipelines (ingestion, automated quality validation, feature generation, real-time inference) and ran distributed multi-GPU / multi-node training (Docker, Kubernetes) to support large-scale experimentation.

NextTier June 2024 - Sept. 2024

Software Engineer Intern - AI Reasoning Agents

Investigated agentic LLM architectures for multi-step query resolution, building a multi-agent system with autonomous reasoning, dynamic tool use (retrieval, web search, code execution), and task decomposition, grounded by a retrieval-augmented generation (RAG) pipeline over domain knowledge.

Designed an evaluation framework measuring task success rate, tool-call efficiency, failure-recovery rate, and multi-step completion accuracy, enabling systematic comparison of agent configurations and quantifying gains in factual grounding.

SELECTED PROJECTS

Residual Network - ResNet-50 Implemented from Scratch View Code

Personal Project • TensorFlow / Keras

Reconstructed the ResNet-50 architecture (~23.6M parameters) from first principles-custom identity and projection residual blocks with skip connections and batch normalization-to probe how residual connections

shape optimization and gradient flow in very deep networks.

Trained and evaluated on a six-class hand-sign image-classification task, reaching ~87.5% test accuracy and empirically confirming the optimization advantage of residual learning over plain deep stacks.

Stock-Price Forecasting with Stacked LSTMsView Code

Personal Project • TensorFlow / Keras

Studied sequence modeling for noisy financial time series, building a stacked four-layer LSTM (50?60?80?120 units) with dropout regularization to forecast next-step prices for Google (GOOG) and Oracle (ORCL) from multivariate OHLCV inputs.

Engineered the sequence-windowing and normalization pipeline and trained per-ticker models under a mean-squared-error objective, comparing model variants to assess generalization across the two equities.

RELEVANT COURSEWORK

Machine Learning & AI: Machine Learning & Artificial Intelligence (A?)

Mathematics & Statistics: Linear Algebra (A), Real Analysis (A), Probability Theory (A), Linear Programming (A?), Matrices (A), Multivariable & Vector Calculus (A), Differential Equations (A?), Mathematical Statistics, Discrete Mathematics, Graph Theory

Computer Science: Programming Language Concepts (A?), Digital Logic Design (A?), Numerical Analysis I & II, Data Structures & Algorithms, Theory of Computation, Operating Systems, Systems Programming

TECHNICAL SKILLS

Languages: Python, C++, C, Java, MATLAB

ML / DL: PyTorch, TensorFlow, Keras, MMDetection, OpenCV, Scikit-learn, NumPy, Pandas; CNNs, 3D-CNNs, LSTMs; object detection (YOLO, YOLO-World), knowledge distillation, continual learning

Systems / Tools: Linux, Docker, Kubernetes, Git, FastAPI, PostgreSQL, Jenkins